AirSculpt

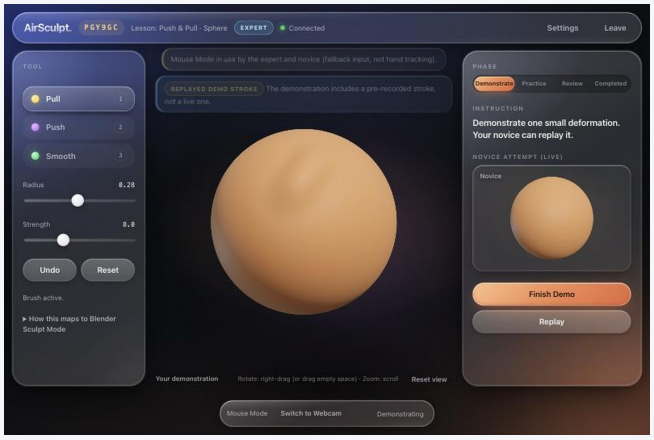
Remote 3D sculpting, taught through movement

CPSC 581 | Group Project 2 | Expertise Sharing

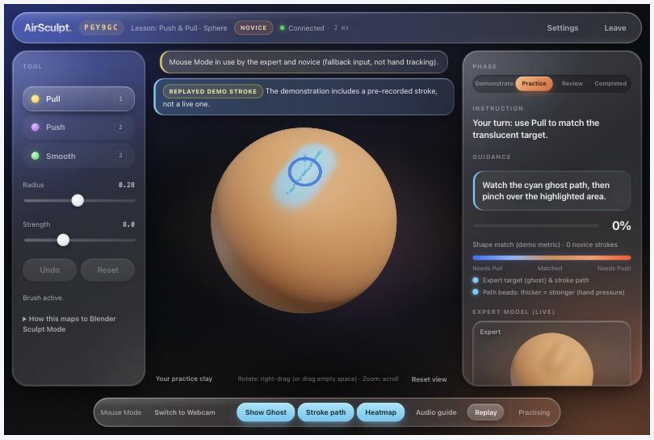
Team members

Hardik Mehta | Shashwat Tank | Happy Prajapati | Jawad Naheen Shamim

EXPERT TEACHES



NOVICE PRACTISES



Final studio views: the expert demonstrates and the novice independently sculpts with guidance.

Concept

AirSculpt makes expert 3D sculpting techniques shareable through hand movement. The expert's stroke becomes a live translucent ghost, the novice sculpts independently and receives directional geometric feedback. A whole-class mode extends the same teaching loop to student phones.

Application: https://airsculpt.onrender.com

1. Project overview

AirSculpt is a synchronous, remote 3D sculpting learning. An expert demonstrates Blender Sculpt Mode inspired techniques like Pull, Push and Smooth on a 3D clay object. A novice sees the demonstration represented directly in the same geometric workspace as a translucent ghost and timed stroke path, then applies the technique to a separate model. The system compares the models and gives the novice specific corrective feedback.

The main challenge is not simply making a remote 3D editor. It is conveying the tacit parts of sculpting expertise where to place the brush, which direction to deform the surface, how strongly to sculpt, and how to recognise and correct a mismatch. AirSculpt uses webcam tracked finger movement and pinch as embodied input, then communicates motion, pressure and correction through geometry, colour, animation, pointing and optional audio.

The original system supports a private expert–novice pair. A later iteration adds whole-class teaching, in which students join by QR code and independently sculpt on their phones while the expert sees a live class wall.

2. Design goals

- Communicate sculpting movement in 3D space, not only an image of the expert's desktop.
- Keep expert demonstrations and novice attempts independent, while making their differences visible.
- Support embodied participation with webcam finger tracking and direct touch, alongside reliable mouse input.
- Make correction actionable; show where more material is needed, where excess must be removed and whether progress improves.
- Let the expert teach one person or a classroom without making participants manage complex CAD software.
- Keep the clay model central, with clear system status, unobtrusive guidance and dependable controls.

3. Design process

1 DIVERGE	2 COMPARE	3 PROTOTYPE	4 TEST	5 REFINE	6 EXTEND

The design process is presented as divergent exploration followed by convergence, prototyping, evaluation and refinement. AirSculpt's repository contains six actual tagged implementation iterations and annotated screenshots. The pencil sketches document alternative concepts and interaction decisions; where a sketch or comparison was created retrospectively to explain the work, it should not be mistaken for a contemporaneous usability-study record.

3.1 Methods used to analyse and document the design

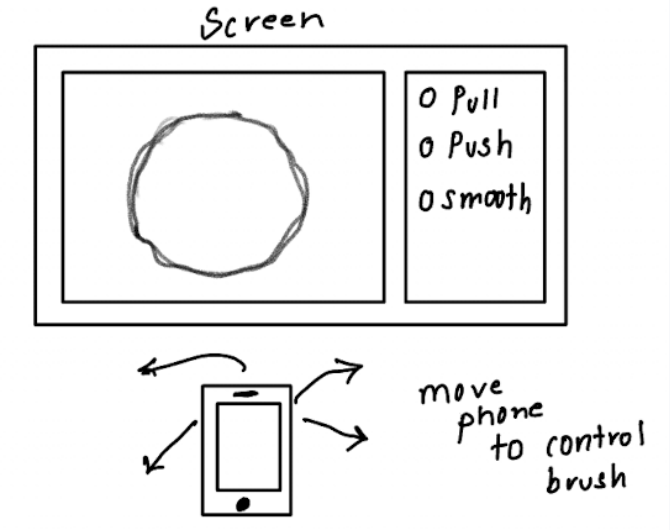
Method	Why appropriate	Design contribution
Task decomposition	Break a sculpting lesson into place, direction, force and correction.	Motivated pinch control, pressure, demonstration traces and directional hints.
Low-fidelity pencil sketching	Explore embodied inputs and interface layout without coding investment.	Phone wand, webcam pinch, stylus, expert/novice wireframes and feedback concepts.
Scenario storyboard	Show roles and information flow across a remote lesson.	Kept expert demonstration, novice imitation and correction as one loop.
Qualitative concept matrix	Compare candidate ideas against assignment-specific criteria.	Justifies accessible webcam tracking with a mouse fallback.
Iterative prototyping and walkthrough	Observe real consequences of implementation decisions.	Six versions; UI, feedback, geometry and classroom refinements.
Heuristic review and task-based testing	Examine feedback, error prevention, navigation and completion.	More visible system state, two-click reset, safer input handling and clearer view rotation.

4. Divergence - exploring embodied input

The first design question was what physical action could convey a sculpting demonstration naturally. Three pencil concepts are shown below. They represent alternatives with different hardware, effort and instructional trade-offs.

SKETCH 1

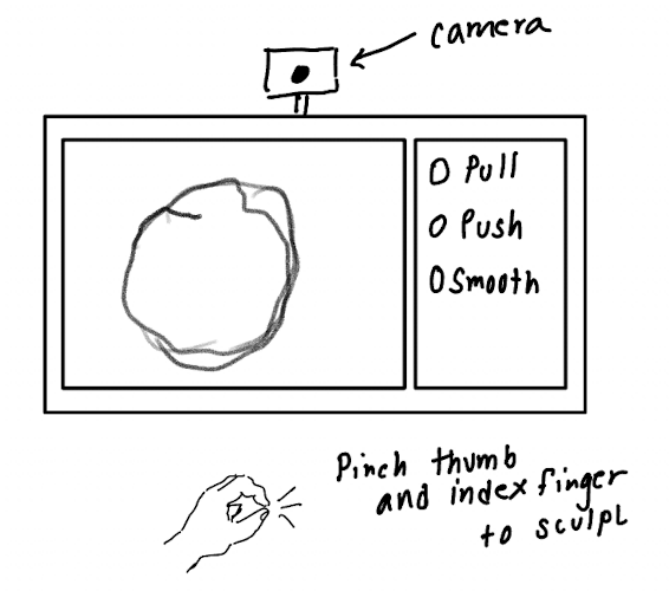
Phone as sculpting wand



Moving a phone through space could steer a virtual brush, tilt and touch might alter the action.

SKETCH 2

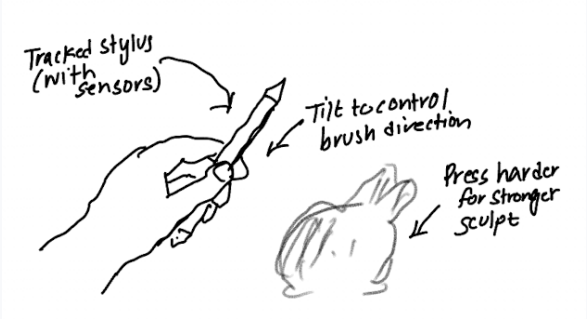
Webcam hand-pinch sculptor



Point with the index finger and pinch the thumb/index to begin and end a stroke.

SKETCH 3

Tracked stylus concept



A pressure sensitive pen could provide fine control but needs additional hardware.

Each concept makes movement central instead of relying on screen sharing. The phone wand is expressive but introduces device alignment and pose-tracking uncertainty. The stylus supports precision but raises equipment requirements. The webcam approach turns the user's own hand into the teaching instrument while preserving a straightforward laptop-only path.

5. Convergence - selecting a practical interaction

5.1 Qualitative comparison of alternatives

Candidate	Embodiment	Teaching value	Practical constraint / conclusion
Phone wand	Strong spatial movement	Could demonstrate sweeping gestures.	Needs a separate device and stable mapping between phone motion and model coordinates not selected.
Webcam pinch	Direct hand tracking	Communicates cursor position, stroke timing and pressure using a visible gesture.	No dedicated hardware beyond a webcam chosen as primary expert input.
Tracked stylus	Fine hand control	Strong for detailed stroke precision.	Additional tracking/pen hardware reduces accessibility not selected.
Mouse / touch	Less expressive but familiar	Clear fallback for practice and reliable classroom joining.	Retained as an intentional parallel input, not a replacement for the embodied concept.

5.2 Why webcam pinch plus a mouse fallback?

The webcam requires no specialised sensing accessory MediaPipe hand landmarks locate the index fingertip and detect the thumb index pinch. The index finger sets the brush location, and pinching makes a deliberate distinction between pointing and editing. Tool selection stays explicit in the interface the system does not need to recognise multiple ambiguous hand poses for Pull, Push and Smooth. A calibrated hand-distance signal can modulate brush strength, giving the expert another expressive channel.

Mouse Mode is deliberate error resilience. It preserves the same sculpting semantics when a camera is unavailable, permission is denied, lighting is poor or a classroom demonstration needs a predictable backup. The system labels Mouse Mode rather than presenting it as hand tracking.

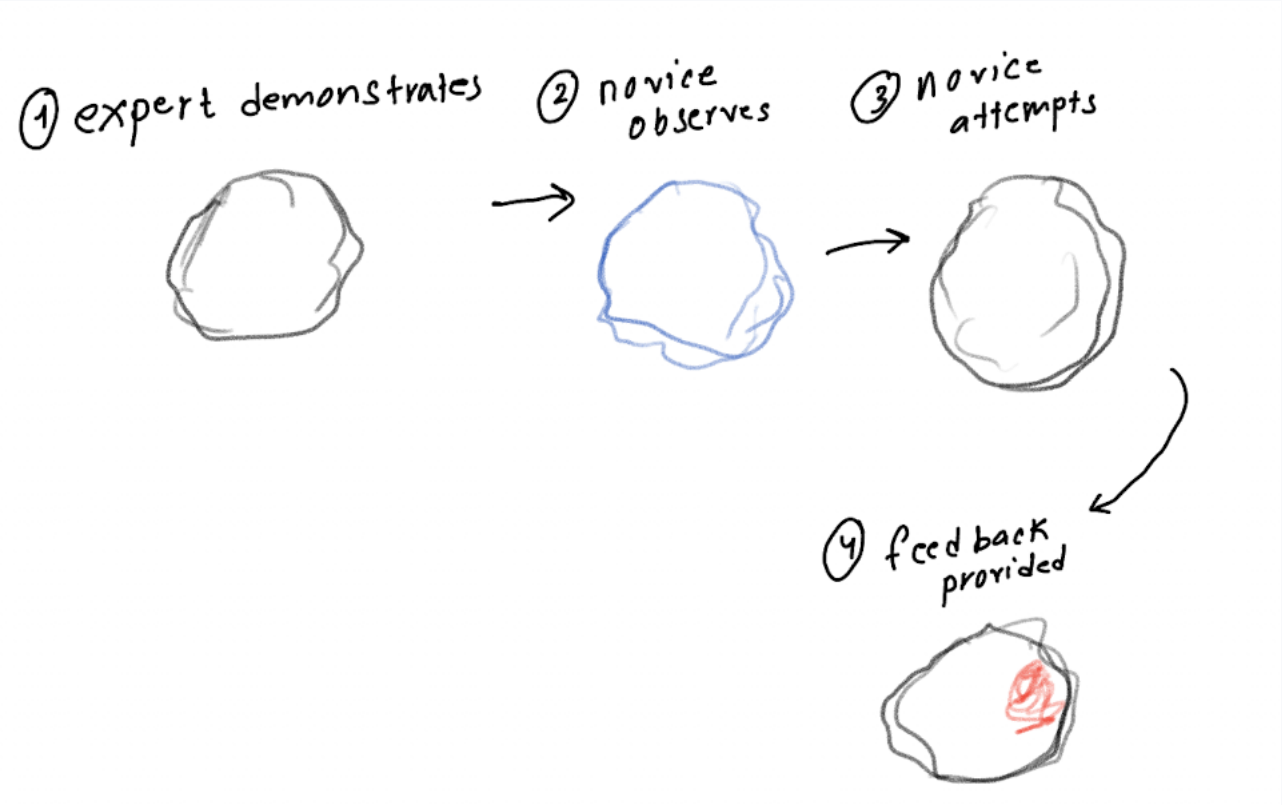
> **Central interaction decision**
> Keep hand movement and gesture as the distinctive teaching input, but do not make a successful lesson dependent on perfect hand recognition.

6. Learning scenario - expert to novice

The teaching cycle is a sequence of observable actions rather than a general-purpose meeting. The expert chooses an object and demonstrates a small deformation, the novice receives the live ghost, practises on an independent copy, reads actionable feedback, and optionally requests or receives a correction pass.

SKETCH 4

Expert–novice storyboard



Four panels: expert demonstrates, novice observes ghost, novice attempts and system shows correction.

6.1 Task analysis

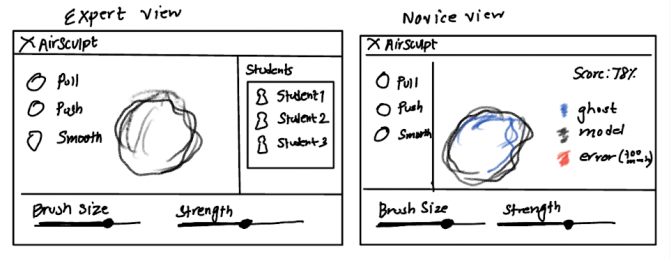
Learning question	Information embodied in the interface	Why it matters
Where should I sculpt?	Tracked fingertip / cursor plus ghost path and expert pointer.	Makes the demonstration spatial, not merely verbal.
When does a stroke start or end?	Pinch-open transitions or mouse press/release; visible brush feedback.	Supports deliberate gestures and avoids accidental edits.
In which direction?	Pull or Push tool, signed heatmap and corrective ring/arrow.	Turns geometric mismatch into a next action.
How strongly?	Depth-based pressure and strength-sized stroke beads.	Exposes stroke intensity that a flat screenshot would hide.
How close is my result?	Region-aware Shape match and task hints.	Gives formative feedback without falsely calling a blank sphere a success.

7. Low-fidelity interface and gesture decisions

The expert and novice have different responsibilities. The expert controls demonstration, target locking and correction; the novice controls their own sculpture, assistance layers and review. Role separation also prevents the expert's demonstration from silently editing the learner's model.

SKETCH 5

Expert and novice wireframes



Two rough screen rectangles: prominent clay centre, tool strip, expert demo controls, novice feedback.

SKETCH 6

Gesture mapping and fallback



Four drawings: index pointing, thumb index pinch, hand distance, mouse click/drag.

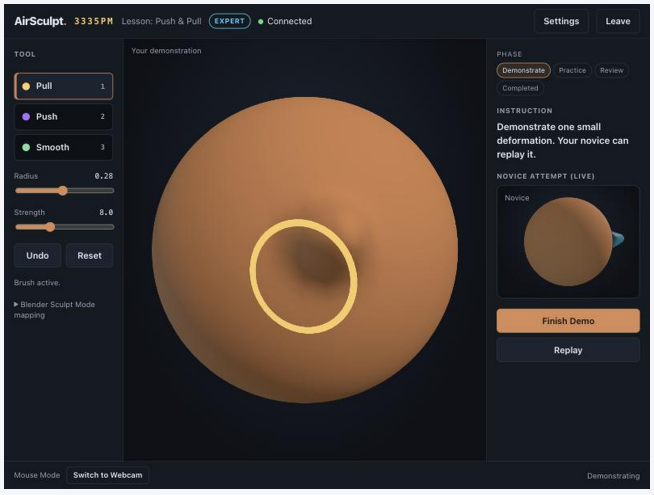
7.1 Translating the sketches into interaction rules

- Finger position maps into the 3D viewport through a projected fingertip and raycast, so the brush sits on the clay surface.
- A stable pinch begins a stroke release, or lost tracking ends it safely, reducing accidental continuous sculpting.
- Pull moves the surface outward, Push moves it inward and Smooth relaxes the local mesh toward its neighbours.
- Undo and restart support safe exploration the expert cannot finish an empty demonstration.
- Zoom, orbit and Reset view keep the 3D form inspectable without confusing camera movement with sculpting.

8. Iteration 1 - establishing the teaching loop

The first functional prototype established a six-character session, explicit expert/novice roles, separate clay meshes, webcam calibration, mouse fallback, streamed strokes, a translucent ghost and lesson phases from demonstration to review. The model and the apprentice's copy start from the same base geometry but are never the same mutable mesh.

EXPERT DEMONSTRATION



NOVICE PRACTICE



Iteration 1: the expert sculpts a target the novice receives a ghost while retaining an independent model.

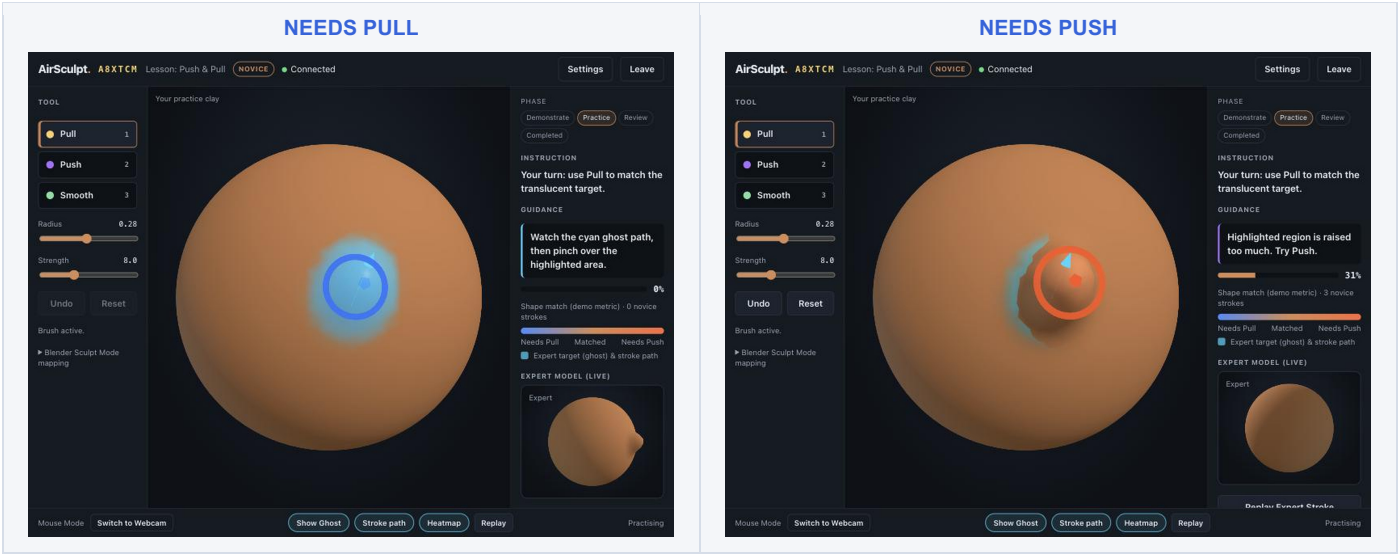
8.1 Findings that drove the next iteration

The original global mean-error score could report approximately 97% similarity when the novice had done nothing, because thousands of unchanged vertices dominated the comparison. A single tint did not communicate whether a region needed more or less material, and expert guidance after the demonstration was limited. These were failures of educational feedback, not just visual polish.

Finding	Design response	Resulting principle
Untouched sphere scored 97%.	Replace global error with comparison focused on the changed task region.	Progress must reflect performance on the taught skill.
Single-colour mismatch was ambiguous.	Introduce signed colour mapping and corrective direction.	Feedback should prescribe an action.
Post-demo expert guidance was passive.	Add live Point & guide on the novice's geometry.	Teaching should remain synchronous during practice.

9. Iteration 2 - making feedback directional

Iteration 2 changed the learner's feedback from a generic percentage to a task-specific correction system. The region-aware score gives 0 to an untouched shape, 100 to an exact copy and lower scores for excessive or misplaced edits. Per-vertex signed colouring identifies insufficient material in blue (Pull) and excess material in orange-red (Push).



Iteration 2: the same clay object visibly distinguishes under-sculpted and over-sculpted regions.

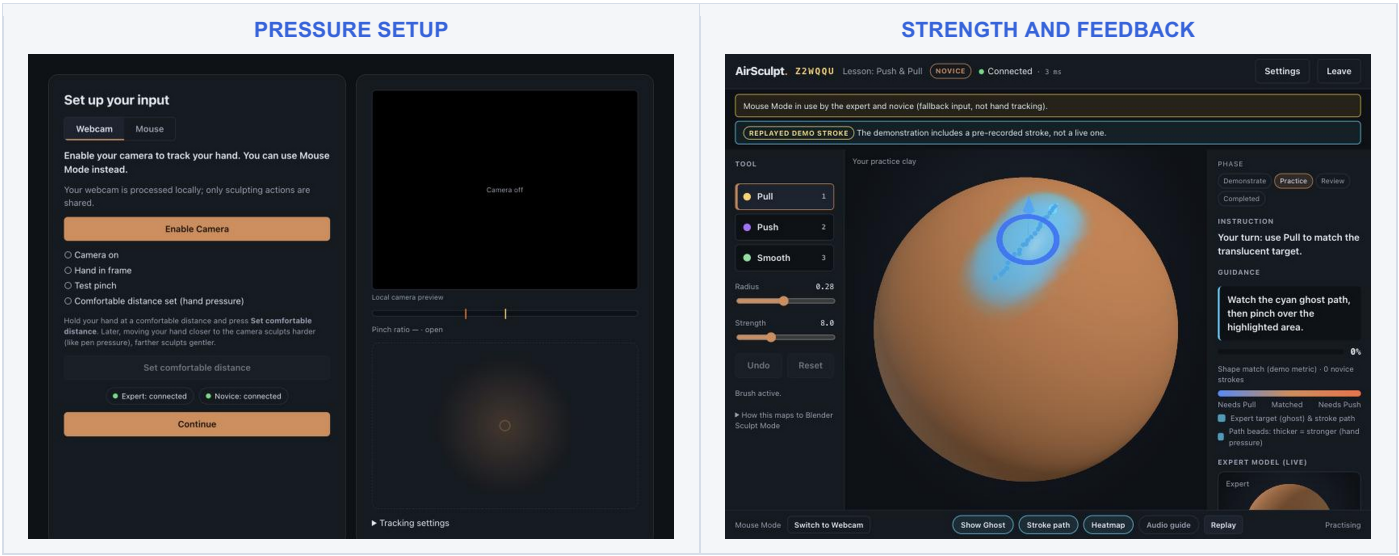
The live Point & guide mode allows the expert to indicate a location on the novice's clay without modifying it. A welded, UV-free icosphere eliminated awkward topology at the seam and poles hints became consistent between the two clients because they update at completed-stroke boundaries.

9.1 Why these changes mattered

For a novice, a score alone answers "how close?" but not "what next?" The paired heatmap, ghost and expert pointer translate a mismatch into a spatial instruction. This is the key embodied-output contribution of AirSculpt guidance is anchored to the object being manipulated rather than located in a remote video or text chat.

10. Iteration 3 - conveying pressure and multisensory feedback

Iteration 3 extended the expertise being transmitted. Hand distance from the camera modulates brush strength from 0.5× to 1.6× after calibration the ghost stroke uses larger and brighter beads for stronger samples. The novice can therefore see the magnitude of the expert's movement as well as its path.



Iteration 3: calibration adds a strength channel and the novice view includes marked demonstration samples.

Optional audio provides a progress-following tone and event cues, without requiring learners to watch a separate dashboard. A study logger, JSON/CSV export and a verbal-only baseline mode were also implemented as evaluation instruments. Their availability is not evidence that a controlled participant study was conducted no participant outcomes are asserted in this report.

10.1 Reliability decisions

A labelled replayed demonstration stroke and visible Mouse Mode state support a predictable classroom demonstration without misrepresenting recorded actions as live tracking. The live connection latency indicator and local landmark overlay help explain what the system is sensing.

11. Iteration 4 - sculpting beyond one sphere

Feedback on the working system exposed two constraints users could not easily discover how to rotate the model, and a sphere-only lesson felt too narrow. The fourth iteration added right drag rotation, drag on empty space rotation, zoom, Reset view and explicit on-screen navigation instructions.

TORUS LESSON



CAPSULE LESSON



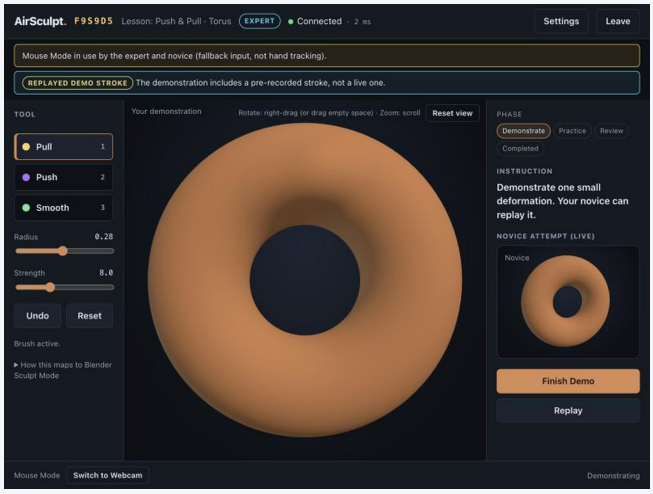
Iteration 4: the same Pull / Push / Smooth teaching flow operates on objects with different surface shapes.

The expert can choose a sphere, rounded cube, capsule or torus before a demonstration. Sculpting and comparisons use each object's base surface normals, making the approach usable on flatter, curved and concave regions. Shape choice is locked once sculpting begins so stored stroke positions remain meaningful.

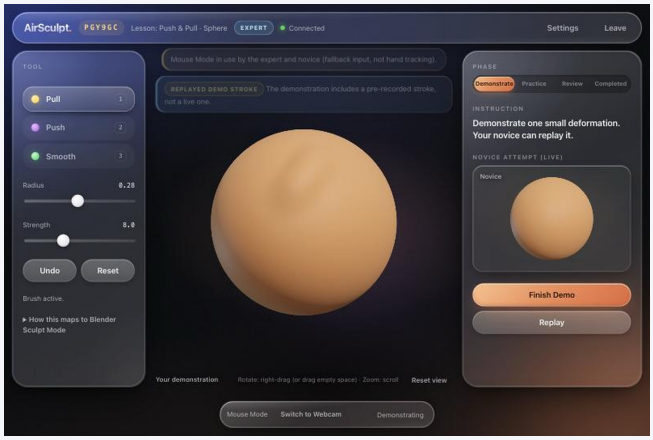
12. Iteration 5 - refining the interface hierarchy

The fifth iteration changed the visual design while preserving the task structure. Earlier panels were functional but visually dense. The refined interface brings the clay forward into a full-bleed stage, with translucent glass panels, clearer control grouping, more balanced lighting and a premium clay material.

EARLIER TOOL LAYOUT



REFINED STUDIO



Before and after: controls become floating supporting surfaces while the sculptable object stays dominant (inspired by iOS liquid glass).

12.1 Evaluation-led refinements

Observed design concern	Refinement	Rationale
Large tool panels competed with the clay.	Floating translucent panels over a spacious 3D stage.	Keep the demonstrated deformation as the perceptual focus.
Over-exposed clay flattened visible relief.	Reduced exposure, sheen and ambient light.	Make subtle ridges easier to read.
Review panes cropped objects on phones.	Stacked review views at narrow widths.	Preserve full geometry comparison.
Rotational interactions were hard to discover.	Keep view instructions and Reset view accessible.	Support recognition rather than memory.

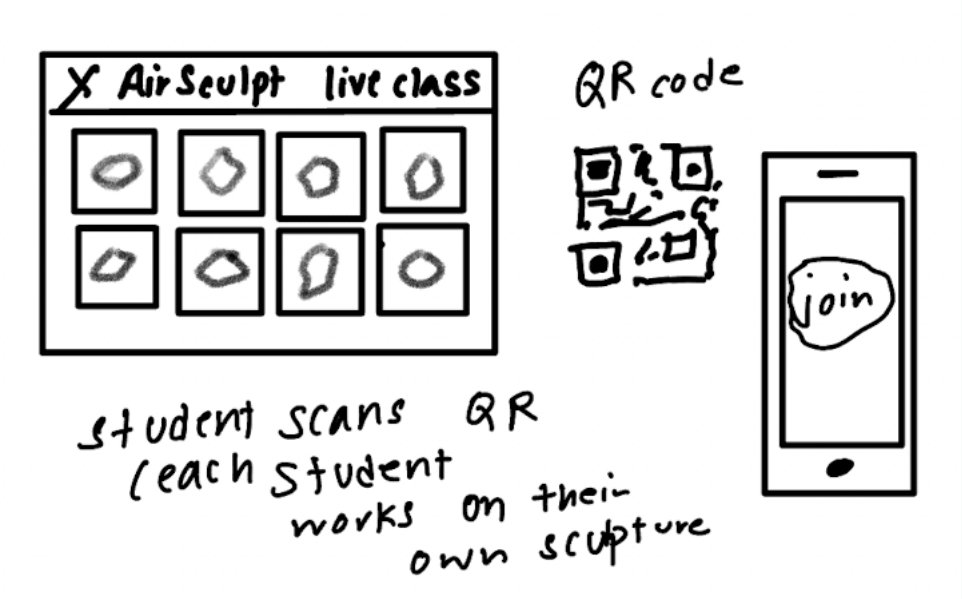
The redesign uses restrained system typography, glass-like surfaces and colour for feedback meaning. It retains focus visible styling and reduced motion handling.

13. Iteration 6 - turning the class into novices

The final major iteration changes the participation model. An expert can open a class session; students scan an on-screen QR code, enter a name and join from their phones without a camera-calibration step. The expert demonstrates once, and each student receives the demonstration on a separate clay model with independent edits, scores and hints.

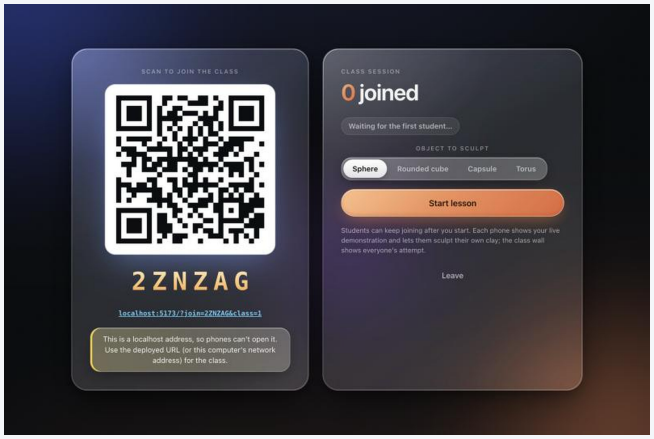
SKETCH 7

Whole-class participation

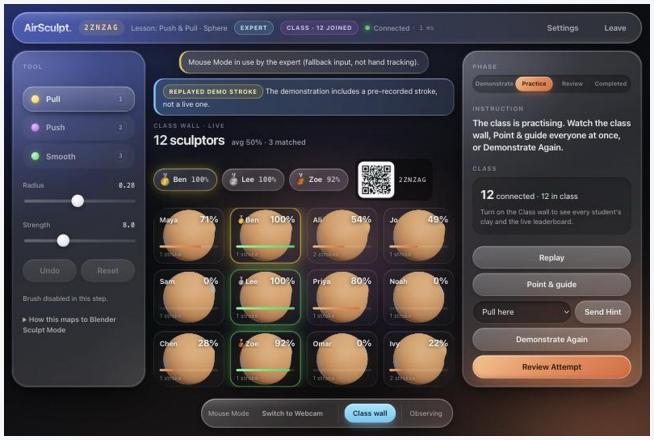


Projector and QR code, students on phones, live class wall with separate student tiles.

JOIN BY QR



LIVE CLASS WALL



Iteration 6: the class lobby invites participation and the teacher sees live independent student progress.

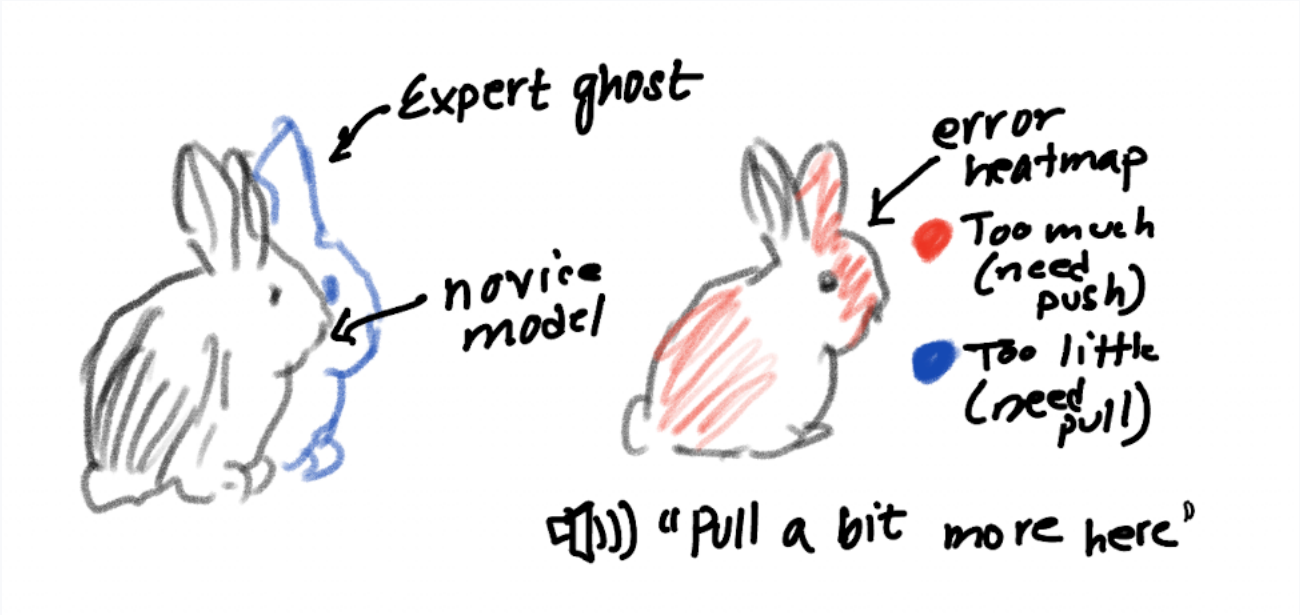
The class wall shows each student's model, match value and progress in one dashboard, while the expert can broadcast Point & guide and hints. The implementation isolates student strokes, so learners do not receive one another's edits.

14. Feedback vocabulary - ghost, heatmap and correction

The visual layers have different purposes. The cyan ghost communicates the expert's target and the stroke path; the region heatmap explains the current difference; the live expert pointer directs attention, and a short hint suggests the next tool. A single layer would not communicate all four dimensions clearly.

SKETCH 8

Ghost and corrective-feedback states



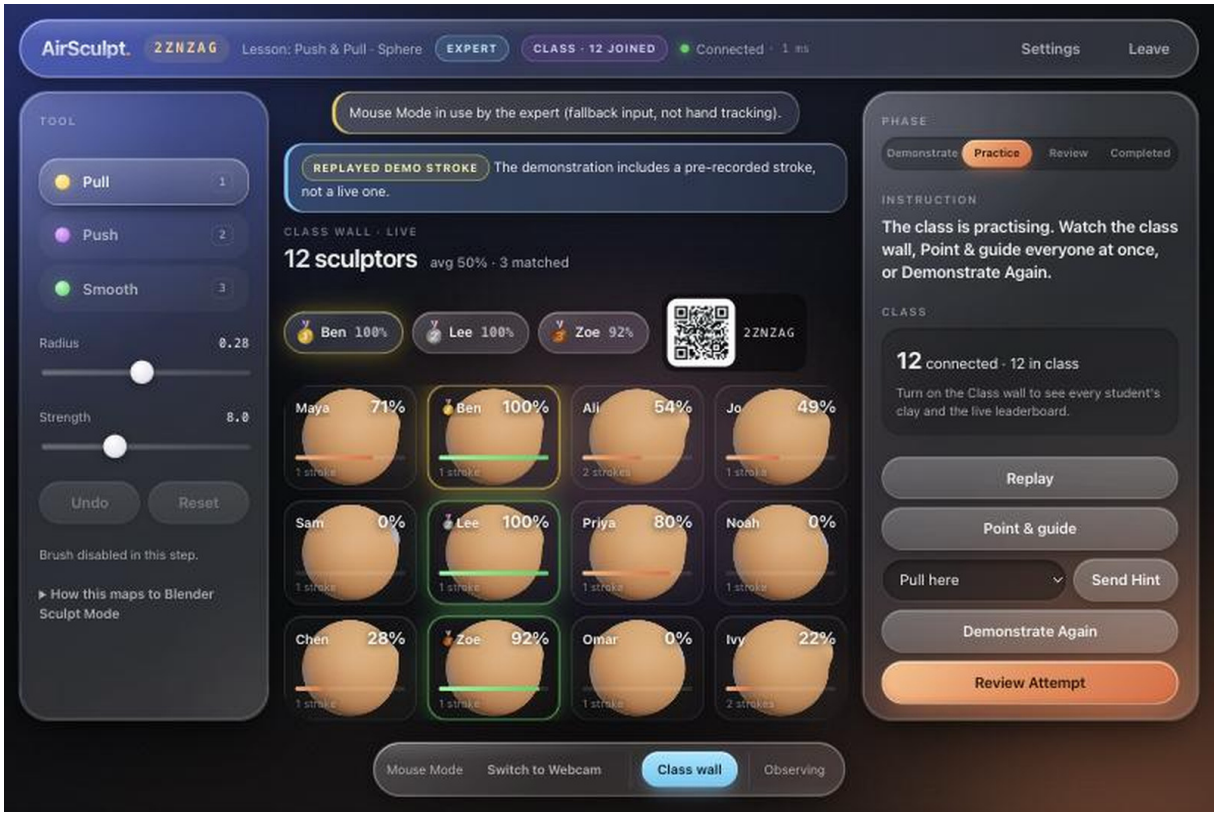
Two clay circles: target ghost over attempt, second circle with marked Pull / Push regions and hint.

Feedback channel	What the novice learns	Design justification
Translucent ghost and stroke beads	Target shape, path and pressure of the demonstration.	Preserves movement information in the 3D object space.
Signed colour heatmap	Blue needs more material, orange-red needs less.	Makes error directional rather than merely numerical.
Shape match and short hints	Whether the attempt improved and which action to try.	Actionable feedback that is easy to check mid task.
Expert pointer / optional audio	Attention location and progress without altering the model.	Supports continued synchronous coaching with multimodal output.

15. Heuristic evaluation

HCI principle	Supporting design decision
Visibility of system status	Current phase, active input mode, score, connection state and clear visual overlays.
Error prevention and recovery	Empty demonstrations blocked, safe tracking-loss release, Undo, Reset view and reconnect handling.
Recognition over recall	Visible tool labels, legend, brush feedback, rotation help and short procedural hints.
Control and flexibility	Camera or mouse, optional ghost/heatmap/audio, independent replay and correction pass.
Aesthetic and minimalist design	Clay-centred stage, simplified tools and secondary supporting panels.

16. Final application and its design rationale



Final classroom demonstration interface: independent student sculptures and a shared progress overview.

AirSculpt extends a one to one, gesture-driven teaching loop to an entire room without changing its conceptual core. A learner still watches a demonstration as spatial data, attempts the motion personally and receives a direction for improvement. Classroom mode changes only the distribution and aggregation of that experience.

Problem	Final design response
Video and screen-sharing omit the physical meaning of a brush stroke.	Transmit position, pinch timing, pressure and resulting geometry as an embodied demonstration.
A novice can see a target but not know what to correct.	Ghost plus signed heatmap, hints and expert spatial pointing.
The expert could accidentally do the learner's work.	Separate meshes, distinct roles and a pointer that never sculpts the novice model.
A single comparison score can be misleading.	Task region scoring with a baseline and off-target penalty.
Input tracking or a classroom device can vary.	Mouse fallback, direct phone touch and explicit calibration/status feedback.
A demonstration can leave most classmates as spectators.	QR join, independent student models and a class wall with progress summaries.

17. Appropriateness of tools and methods

Method / tool	Why it fits AirSculpt	Evidence
Pencil concept sketches	Quick, low-cost examination of three hardware approaches.	Sketches 1–3; compare input expressiveness and feasibility.
Storyboard and wireframes	Communicate actor roles and sequential feedback.	Sketches 4–5; role-based interface structure.
Interaction mapping / task analysis	Link physical gestures to teachable 3D actions.	Sketch 6; pointing, pinching, pressure and fallback.
Qualitative evaluation matrix	Make trade-offs transparent rather than relying on visual preference.	Section 5 compares novelty, teaching value and feasibility.
Heuristic inspection	Check status visibility, error prevention and visual hierarchy.	Sections 12 and 15 connect UI decisions to HCI principles.
Iteration logs / regression tests	Trace changes and check that existing functionality survives.	Tagged iterations 1–6; screenshots and test records.
React, TypeScript, Three.js	Build an interactive, deterministic 3D application in a browser.	Separate 3D views, editable mesh, ghost and class wall.
MediaPipe, Socket.IO	Capture hand-based input and synchronise strokes remotely.	Camera landmark pipeline, sessions, room rules and class broadcasts.
Microsoft Word	Visual, editable process documentation.	Figures, sketches, comparisons and linked rationale.

18. Incomplete functionality, scope limits and handoff

At the final check all implemented features work and no core feature is known to be broken. The following are scope boundaries or methodological cautions, rather than reported failures:

- The system teaches Blender-like sculpting techniques through its own geometry.
- Room/session state is held in server memory rather than a long-term database a server restart starts new sessions.
- Shape match is a task oriented geometric heuristic and the optional comparison-study controls do not by themselves establish participant learning outcomes.

19. Generative AI disclosure

AI-generated material was treated as support rather than an automatic final answer. The team selected the concepts to keep, reviewed the application behavior, adapted the interface, and remains responsible for the submitted work.

Tool	Use in the project
Claude Code	Used to assist with code implementation based on the team's specifications and architectural decisions.
ChatGPT(OpenAI)	Used for rubric review, usability and heuristic feedback, wording refinement, code/interface feedback, report organization, and formatting of the final documentation. The team reviewed and selected which suggestions to incorporate

PROMPT 1 (MVP)

See the blueprint sketches and build the whole system end to end: expert/novice rooms, a sculptable sphere with Pull/Push/Smooth, webcam pinch control with a mouse fallback, live ghost and replay, shape comparison with hints, review and restart.

PROMPT 2 (Correctness and feedback)

Fix the scoring so an untouched sphere no longer scores high (score only the region the expert sculpted). Add a blue/orange heatmap showing where to Pull or Push. Add a live “Point & guide” pointer so the expert can point at the novice’s clay. Switch to a seam-free sphere mesh and fix the issues from the iteration-1 screenshots.

PROMPT 3 (Extras)

Add hand-depth pressure (closer to the camera = stronger), an optional audio guide whose pitch rises as the match improves, a study logger with CSV export, and a “verbal only” baseline mode for comparison. Add a backup pre-recorded demo stroke, a Mouse Mode banner, a hand-landmark overlay, a latency badge and a Blender mapping table.

PROMPT 4 (Rotation and objects)

Make rotating the view easier (right-drag anywhere, a Reset view button) and let the expert choose the object to sculpt: sphere, rounded cube, capsule or torus.

PROMPT 5 (Visual redesign)

The UI looks generic. Redesign it in Apple’s style, minimal but premium, with liquid-glass panels floating over a full-screen 3D view.

PROMPT 6 (Whole-class mode)

Add a feature, everyone scans a QR code, joins on their phone, watches the expert’s demo live and sculpts their own clay, while the projector shows a live “class wall” of everyone’s attempts with scores and a podium. Test it with a simulated class, document it, tag it as iteration 6 and push.

PROMPT 7 (Documentation Formatting)

The content is done and final don't add, cut, or rewrite any of the substance, and don't touch the wording unless it's a typo or grammar error. I just need it reformatted and cleaned up for submission apply consistent headings/section numbering, fix inconsistent spacing, bullet styles, and heading levels.

20. Conclusion

AirSculpt began with a focused educational problem an expert must make the meaning of 3D sculpting movements visible and actionable to a distant novice. The design progressed from a hand-controlled demonstration and ghost model to directional feedback, pressure and audio, more varied objects, a clearer visual hierarchy and ultimately a whole class learning experience. The most important design lesson is that embodied input alone is not sufficient it must be paired with feedback that tells a learner what to change, where, and why.